HAZOP Study Report: Plant X (Spray Drying, Reaction and Solvent Areas)

Deccan Fine Chem Pvt Ltd (fictional) · Revision 2 · March 2024 · Facilitator: R. Iyer · FICTIONAL DEMO DOCUMENT

1. Scope and method

This study covers spray dryer SD-1 (dryer house), reactor R-201 (reaction bay) with its steam utilities from Boiler B-1, and the solvent room with exhaust fan EF-04. Guidewords were applied node by node. Safeguards listed are the independent barriers credited against each consequence. Risk was ranked on a 5x5 matrix.

2. Summary of high-risk scenarios

Three scenarios ranked High: dust explosion in SD-1 (node SD1-2), runaway reaction in R-201 (node R201-1) and solvent vapour fire in the solvent room (node SR1-3). Each relies on several safeguards being available at the same time; the study team noted that simultaneous impairment of two or more safeguards is not covered by the ranking and must be managed through management of change.

Node SD1-2: Spray dryer SD-1, drying chamber. Deviation: MORE temperature

Consequence: ignition of dried powder cloud leading to dust explosion and fire in the dryer house; potential multiple fatalities (Severity 5).

#	Safeguard (independent barrier)	Tag / reference
1	Inlet high-temperature alarm and interlock TAH-301 (trips burner at 180 °C)	TAH-301
2	Batch size kept within validated limit of 400 kg per charge	PV-SD1-2024 Table 3
3	Nitrogen inerting keeps O2 below 8% in chamber	PAL-302
4	Explosion vent panel sized for 400 kg charge dust loading	EV-SD1
5	Only trained operators (dryer competency card) run SD-1	SOP-HR-05

Recommendation SD1-2.1: any bypass of TAH-301 to be treated as a change requiring plant-head approval (see SOP-SD1-03).

Node R201-1: Reactor R-201. Deviation: LESS temperature during batch hold

Cause: loss of jacket steam from Boiler B-1. Consequence: batch cools below 55 °C; dosed reagent accumulates unreacted and can react violently when heating resumes (runaway, Severity 5).

#	Safeguard (independent barrier)	Tag / reference
1	Jacket temperature control using steam from Boiler B-1	TIC-201 / B-1
2	Reactor high-temperature alarm TAH-201	TAH-201
3	Batch size within validated limit	PV-R201-2023
4	Only trained operators run R-201	SOP-HR-05

Recommendation R201-1.2: maximum allowable steam interruption during batch hold to be defined in the operating SOP (implemented as SOP-R201-07 §4.2: 20 minutes).

Node SR1-3: Solvent room. Deviation: LESS ventilation

Consequence: toluene vapour accumulates towards its lower explosive limit; ignition from electrical equipment or sparks leads to fire or explosion (Severity 5).

#	Safeguard (independent barrier)	Tag / reference
1	Exhaust fan EF-04 delivering at least 12,000 m3/h	EF-04
2	Gas detector GD-07 alarms at 20% LEL	GD-07
3	Ex-rated electrical equipment for Zone 1, IIB, T4	AC-SR1
4	Hot-work permit control	SOP-MNT-02

Recommendation SR1-3.2: replacement of EF-04 must be like-for-like on airflow, Ex certification, temperature class and spark-resistant impeller (see AC-SR1, PS-MECH-11).